Compiling Facts into Applications

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We define AREST, a compositional architecture unifying Backus's Formal Functional Programming and Applicative State Transition systems [1] with Fielding's Representational State Transfer [4], using Halpin's ORM 2 [5] as the surface syntax. Domain knowledge is expressed as FFP objects in FORML 2 [6]; the representation function ρ maps each object to its executable; the system state is Backus's AST cell sequence; the persistent data is Codd's named set [3]. We prove eight properties: grammar unambiguity, specification equivalence, completeness of state transfer, HATEOAS as projection, derivability, query adequacy, derivation completeness, and scope.

CCS Concepts: • **Software and its engineering** → **Formal language definitions**; **Domain specific languages**; • **Theory of computation** → *Constraint and logic programming*; • **Information systems** → *RESTful web services.*

Additional Key Words and Phrases: applicative state transfer, REST, formal functional programming, object-role modeling, state machines, HATEOAS, first-order logic, specification equivalence

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1 Definitions

Definition 1.1 (AREST). An AREST system is a tuple (O, ρ, D, P, sub) where:

- O is the set of FFP objects [1, Sec. 13.2]
- $\rho : O \to (O \to O)$ is the representation function [1, Sec. 13.3.2]
- D is the AST state, a sequence of cells [1, Sec. 14.3]
- $P = \uparrow FILE : D$ is the population, a named set of relations [3, Sec. 1.6] whose elements are ORM 2 elementary facts [5, Ch. 2]
- sub is the subsystem function dispatching inputs to state transitions, producing output as REST representations [4] with HATEOAS links

Three terms carry overloaded abbreviations. *AST* follows Backus's usage (Applicative State Transition, 1978), not the compiler term. *ORM* follows Halpin's usage (Object-Role Modeling, 2008), not the persistence pattern. *FP* follows Backus's usage (Function-level Programming), not the broader tradition.

The primitive atoms of O include Backus's structural functions [1, Sec. 11.2.3]. Sequences represent functional forms via metacomposition [1, Sec. 13.3.2]:

$$(\rho\langle x_1, \ldots, x_n\rangle) : y = (\rho\, x_1) : \langle\langle x_1, \ldots, x_n\rangle, y\rangle \tag{1}$$

The first element determines which functional form; the remaining elements are its parameters. In AREST, a FORML 2 reading is an object; ρ extracts the constraint or derivation function it represents.

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2 Operations

2.1 Domain Primitives

Each combining form [1, Sec. 11.2.4] is an FFP object with a domain interpretation:

Table 1. Backus combining forms as FFP objects with AREST domain interpretations.

FFP Object	Form	Domain
$\langle CONS, s_1, \ldots, s_n \rangle$	Construction	Graph Schema
s_i	Selector	Role
$\langle COMP, f, g \rangle$	Composition	Derivation
$\langle COND, p, f, g \rangle$	Condition	Constraint
$\langle \alpha, f \rangle$	Apply-to-all	Traversal
$\langle /, f \rangle$	Insert	Aggregation
$\langle bu, f, x \rangle$	Binary-to-unary	Query
$\bar{x}$	Constant	Literal

A graph schema $\langle CONS, s_1, \ldots, s_n \rangle$ applied to all roles is a fact; partially applied via *bu* it is a query.

2.2 Relational Algebra

Codd's adequate θ_1 [3, Sec. 2.2] as FFP definitions:

$$\pi_L(R) \equiv \alpha\,[s_{i_1}, \ldots, s_{i_k}] : R \tag{2}$$

$$R * S \equiv Filter(eq \circ [s_{sh}]) \circ distl \tag{3}$$

$$R_{L|_M} S \equiv Filter(p) : R \tag{4}$$

$$R \cdot S \equiv \pi_{1s}(R * S) \tag{5}$$

$$\gamma(R) \equiv Filter(eq \circ [s_1, s_n]) : R \tag{6}$$

2.3 State and Cells

The state D is a sequence of cells $\langle CELL, n, c \rangle$ [1, Sec. 14.3]:

$$\uparrow n : D \longrightarrow c \qquad \text{(fetch)} \tag{7}$$

$$\downarrow n : \langle x, D \rangle \longrightarrow D' \qquad \text{(store)} \tag{8}$$

FILE is a cell whose contents is P. Each entity is a cell. Cells may contain sub-stores [1, Sec. 14.7]; partitioning is a cell containing a subset of P.

2.4 Subsystem Dispatch

Following [1, Sec. 14.4.2]:

$$\begin{aligned}
&sub \equiv \\
&\quad is\text{-}chg \circ \uparrow K \longrightarrow [rpt, apl] \circ [\uparrow I, defs]; \\
&\quad is\text{-}qry \circ \uparrow K \longrightarrow [qry \circ [\uparrow I, \uparrow F], defs]; \\
&\quad is\text{-}cmd \circ \uparrow K \longrightarrow chk \circ apl \circ [\uparrow I, defs]; \\
&\quad is\text{-}upd \circ \uparrow K \longrightarrow \\
&\qquad [rpt, \downarrow F \circ [upd, defs]] \circ [\uparrow I, \uparrow F]; \\
&\quad [err \circ [\uparrow K, \uparrow I], defs]
\end{aligned} \tag{9}$$

where $K = KEY$, $I = INPUT$, $F = FILE$.

2.5 Commands

Each interaction is an AST transition [1, Sec. 14.3.1]:

$$\mu(SYSTEM{:}x) = \langle o, D' \rangle \tag{10}$$

A Create command:

$$create = emit \circ validate \circ derive \circ resolve \tag{11}$$

A state machine is a fold [1, Sec. 11.2.4]:

$$machine = /\,transition : \langle e_1, \ldots, e_n \rangle \tag{12}$$

$$transition = (p \longrightarrow \bar{s'};\ id) \tag{13}$$

3 Example

An order domain is declared in FORML 2:

```
Order(.OrderId) is an entity type.
Customer(.Name) is an entity type.
Order was placed by Customer.
  Each Order was placed by exactly one
  Customer.
Status 'Draft' is initial in State Machine
  Definition 'Order'.
Transition 'place' is from Status 'Draft'.
  Transition 'place' is to Status 'Placed'.
```

The constraint reading is an FFP object; ρ yields:

$$(\rho\ reading) = \langle COND,\ (count \leq 1) \circ Filter(eq \circ [s_1, \bar{o}]),$$
$$\bar{\phi},\ violation \rangle : P_{placedBy}$$

`POST /orders {"customer":"acme"}` dispatches as *is-cmd* in (9). The engine applies *create* (11):

$$P' = P \cup \{Order(\texttt{ord-1}),\ placedBy(\texttt{ord-1}, \texttt{acme}),$$
$$SM(\texttt{ord-1}, \texttt{Draft})\}$$

Response (Theorem 4.4: links from P'):

```
{ "id": "ord-1", "customer": "acme",
  "status": "Draft",
  "_links": {
    "place": {
      "href": "/orders/ord-1/transition",
      "method": "POST" } } }
```

Following "place" applies (12). The new representation contains $\{ship\}$, not $\{place\}$. If a second command assigns a different Customer to the same Order, the UC fires and the command is rejected. The violation is returned as the original reading.

4 Properties

Theorem 4.1 (Grammar Unambiguity). *Let N be a declared set of object type names and F a declared set of predicate readings, such that no element of N contains a formal item of the grammar as a substring. Then each FORML 2 sentence over (N, F) has exactly one parse.*

Proof. A FORML 2 sentence has the structure:

$$[\mathit{modal}] \quad [\mathit{quant}_1] \quad n_1 \quad r \quad \mathit{quant}_2 \quad n_2 \quad [\mathit{qual}]$$

where $\mathit{modal} \in$ {"It is obligatory that", "It is forbidden that", "It is permitted that", ε}; $\mathit{quant}_1 \in$ {"Each", "For each", "the same", ε}; $n_1, n_2 \in N$; $r \in F$; $\mathit{quant}_2 \in$ {"at most one", "some", "exactly one", "at least k and at most m", "more than one", "no", ε}; $\mathit{qual} \in$ {"who ...", "that ...", ε}.

(1) The modal prefixes are pairwise non-overlapping strings, so *modal* is determined by the sentence prefix.

(2) After *modal*, the quantifier quant_1 is determined: "Each" vs. "For each" vs. "the same" vs. a name from N are distinguishable because no $n \in N$ contains a formal item as a substring (by hypothesis).

(3) n_1 is determined by longest-first matching against N. Since names do not contain formal items, the boundary between n_1 and r is unambiguous.

(4) r is the maximal token sequence between n_1 and quant_2. The quantifier phrases in quant_2 are pairwise distinct strings: {"at most one", "some", "exactly one", "at least", "more than one", "no"}. None is a prefix of another in this position. Since no $n \in N$ contains these phrases, the boundary between r and quant_2 is unambiguous.

(5) n_2 is determined by longest-first matching against N after quant_2.

(6) *qual* is determined by "who" or "that" (personal vs. impersonal pronouns per [7, Sec. 1.5]).

Each structural position admits exactly one parse. The constraint type is determined by the pair $(\mathit{quant}_1, \mathit{quant}_2)$, which maps injectively to constraint kinds: (Each, at most one) $\to$ UC; (Each, some) $\to$ MC; (Each, exactly one) $\to$ UC+MC; (Each, at least k...) $\to$ FC; (the same, more than one) $\to$ UC$^-$. No two pairs map to the same kind. □

Theorem 4.2 (Specification Equivalence). *Let $\mathit{parse} : R \to \Phi$ and $\mathit{compile} : \Phi \to O$ be injective, with R the set of FORML 2 readings, Φ the set of FOL formulas, and O the set of FFP objects. Then:*

$$\mathit{parse}^{-1} \circ \mathit{compile}^{-1} \circ \mathit{compile} \circ \mathit{parse} = \mathit{id}_R \tag{14}$$

and $\rho(\mathit{compile}(\mathit{parse}(r)))$ is the executable for every $r \in R$.

Proof. *parse* is injective by Theorem 4.1: each sentence has one parse, so distinct sentences yield distinct formulas. *compile* is injective: the $(\mathit{quant}_1, \mathit{quant}_2)$ pair determines the constraint kind (Theorem 4.1), which determines the FFP object structure (Condition for UC, null-test for MC, count for FC). Both have left inverses; composing: $\mathit{parse}^{-1} \circ \mathit{compile}^{-1} \circ \mathit{compile} \circ \mathit{parse} = \mathit{id}_R$. Executability: ρ maps the compiled object to a function by [1, Sec. 13.4]. □

Theorem 4.3 (Completeness of State Transfer). *create (11) produces a complete $\langle o, D' \rangle$ containing entity, state machine instance, derived facts, violations, and HATEOAS links.*

Proof. Let $P = \uparrow \mathit{FILE} : D$. (1) *resolve* inserts entity facts: $P' = P \cup E$. (2) *derive* is monotonic over finite P; by Knaster-Tarski, a unique least fixed point P'' exists. (3) *validate* evaluates all constraint objects in D against P'': $V = \{(\rho\, c) : P'' \mid c \in D_{\mathit{constraints}}\}$. (4) *emit* produces $\langle o, D[\mathit{FILE} \mapsto P''] \rangle$ if no alethic violation, else $D' = D$. The state does not change during evaluation [1, Sec. 14.1]. □

Theorem 4.4 (HATEOAS as Projection). *Let $T = \{(s_i, s_j, e_k)\} \subseteq P$ be transition facts and s the current status. Then:*

$$links(s) = \pi_{event}(Filter(p) : T) \tag{15}$$

where $p(t) = (s_{from}(t) = s) \vee anc(s_{from}(t), s)$.

Proof. $T \subseteq P$. *Filter* is restriction (4); π is projection (2). Both are θ_1 operations on the named set. The representation renders P; $links(s)$ is derived from P by θ_1 operations and therefore appears in the representation without external computation. □

Theorem 4.5 (Derivability). *For every domain value v in an AREST representation:*

$$\exists f \in O.\ v = (\rho\, f) : P \tag{16}$$

Proof. Domain values partition into: (1) entity attributes (facts in P; $v = (\rho\, s_i) : fact$), (2) derived facts ($v = (\rho\, r) : P$ for derivation rule r), (3) state machine status (derived fact), (4) links ($v = (\rho\, \langle COMP, \pi, Filter(p)\rangle) : T$ by Theorem 4.4), (5) violations ($v = (\rho\, c) : P$ for constraint c). Each case yields (16). □

Theorem 4.6 (Query Adequacy). *Every query on P is expressible using θ_1, each element of which is an FFP definition (2)–(6).*

Proof. P is a named set of relations in normal form (ORM 2 elementary facts are 5NF by construction [5, Ch. 12]). θ_1 is adequate for the named set [3, Sec. 2.2]. Each element of θ_1 is an FFP definition by (2)–(6). □

Theorem 4.7 (Derivation Completeness). *Every θ_1-derivable relation has a FORML 2 reading. Forward chaining evaluates the corresponding FFP objects to the least fixed point.*

Proof. Each θ_1 operation has a FORML 2 verbalization: projection verbalizes as role selection ("X has Y"); join as shared-role composition ("X who R_1 Y and R_2 Z"); restriction as constraint predicate ("X where Y exceeds Z"); tie as self-reference ("X is ancestor of X"). By Theorem 4.1, each verbalization has a unique parse, so each θ_1 composition has a unique reading. Forward chaining applies derivation rules (FFP objects) to P monotonically; by Knaster-Tarski the least fixed point exists and is reached in $|P_{\max}|$ steps. □

Theorem 4.8 (Scope). *The architecture covers all values that are both factual and computable. Values outside are either non-factual or non-computable.*

Proof. Any factual proposition (a statement with a truth value) is expressible as a tuple in a relation [3, Sec. 1.3], hence a fact in P. Any computable function (an effective method) is expressible as a λ-term [2]; Backus shows the equivalence of FP combining forms to λ-calculus combinators [1, Sec. 12.9], and FFP extends FP with metacomposition. Therefore every factual-and-computable value is $(\rho\, f) : P$ for some $f \in O$, $P \subseteq D$. Non-factual values (no truth value) cannot be relations. Non-computable values (no effective method) cannot be computed by any system. □

Corollary 4.9 (Algebraic Optimization). *Programs composed from FFP combining forms admit algebraic transformation via the laws of [1, Sec. 12.2]. Compilation from FFP objects to optimized evaluation preserves ρ-semantics.*

Proof. The 24 laws of the FP algebra [1, Sec. 12.2] are equations between functional forms. An FFP object representing a program can be transformed to an equivalent object by applying these laws. The ρ-image is preserved because the laws hold for all objects in the domain. Ahead-of-time compilation from objects to evaluation trees is one such transformation. □

5 Conclusion

AREST composes FFP/AST [1] with REST [4] via ORM 2 [5]. Readings are FFP objects; ρ maps them to executables; the population is Codd's named set [3] stored as Backus's FILE cell; transitions produce hypermedia. The eight results establish: the grammar is unambiguous (Theorem 4.1); specifications and executables are the same object (Theorem 4.2); state transfer is complete in one application (Theorem 4.3); HATEOAS is a θ_1 projection (Theorem 4.4); all domain values are derivable (Theorem 4.5); all queries are expressible (Theorem 4.6); all derivations are verbalizable (Theorem 4.7); and the scope is factual-and-computable, which is everything (Theorem 4.8).

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